05 Large intestine, rectum and anal canal

Team: _____

Basic function (this is all you need): Water and salts are absorbed, feces are formed and stored, and defecation is controlled.

STRUCTURES TO DRAW AND LABEL

- Cecum with the ileocecal valve and the vermiform appendix
- Ascending colon (retroperitoneal), right colic (hepatic) flexure
- Transverse colon (intraperitoneal, on the transverse mesocolon), left colic (splenic) flexure
- Descending colon (retroperitoneal), sigmoid colon (on the sigmoid mesocolon)
- Rectum with its transverse rectal folds; anal canal
- Three features seen from outside: teniae coli (three longitudinal muscle bands), haustra (pouches), omental (epiploic) appendices (fat tags)
- Anal canal: anal columns, anal sinuses, pectinate line, internal anal sphincter (smooth muscle), external anal sphincter (skeletal muscle)

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) to the cecum, ascending colon and about the proximal two thirds of the transverse colon; pelvic splanchnic nerves (S2 to S4) to the rest, including the rectum
- Sympathetic: lesser and least splanchnic nerves to the superior mesenteric ganglion; lumbar splanchnic nerves to the inferior mesenteric ganglion
- External anal sphincter: pudendal nerve (somatic, S2 to S4). Internal anal sphincter: autonomic
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Superior mesenteric artery (ileocolic, right colic, middle colic) to the proximal two thirds of the transverse colon; inferior mesenteric artery (left colic, sigmoid, superior rectal) beyond that

THE WALL UNDER THE MICROSCOPE

- Simple columnar epithelium with many goblet cells; no villi; no circular folds
- Deep, straight intestinal glands (crypts) packed with goblet cells
- Abundant lymphoid nodules in the lamina propria and submucosa; the appendix is loaded with them
- Muscularis externa: complete inner circular layer; the outer longitudinal layer is gathered into the three teniae coli
- Serosa on intraperitoneal parts; adventitia on the retroperitoneal parts and the lower rectum
- Anal canal: simple columnar above the pectinate line, stratified squamous below it, keratinized at the anus

CELLS AND WHAT THEY MAKE

Absorptive cells (colonocytes)	Absorb water and electrolytes
Goblet cells	Large amounts of mucus to lubricate feces
Enteroendocrine cells	Local hormones
Stem cells (crypt base)	Replace the lining
Paneth cells	Normally absent or very few in the colon (a difference from the small intestine)

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: a scrunchie

The teniae coli are the elastic threads of a scrunchie. They are shorter than the colon, so they gather it into pouches, the haustra. The fat tags hanging off are the omental appendices.

Mnemonic: vagus to two thirds, pelvic to the rest, pudendal to the exit

The vagus reaches to about the proximal two thirds of the transverse colon. Pelvic splanchnic nerves (S2 to S4) take over from there. The pudendal nerve controls the external anal sphincter, the voluntary one. Same split for arteries: superior mesenteric to two thirds, inferior mesenteric to the rest.